

Circuits Are Basis-Dependent Computations: Transplantation Failure and Rescue Across Task-Matched Networks

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Abstract

Mechanistic interpretability typically treats *circuits*—identifiable subgraphs of attention heads and MLP neurons—as the fundamental units of transformer computation. This framing implicitly suggests circuits should be portable: a head that implements induction in one network should, in principle, be transplantable into another network trained on the same task. We show that direct transplantation fails but is not an absolute impossibility: compatibility can be learned by the host or supplied by sufficiently broad residual-stream adapters. Across 20 independently trained small transformers that all converge on Layer-0 induction circuits (Layer 0 is critical in 20/20 seeds; head-level agreement 23.2%, 95% CI [17.4%, 29.5%], $p = 0.002$ vs. uniform null), moving a known-functional critical head into another network’s critical slot drops accuracy from 0.964 to 0.095, indistinguishable from randomized controls. Two-interface Procrustes alignment reaches 0.084 despite improving residual-stream R^2 , and a seven-method head-interface alignment tournament does not rescue the transplant. However, two rescue experiments change the interpretation: freezing the transplanted donor head and retraining the host recovers $\geq 80\%$ of recipient baseline in 100–500 steps across five seed pairs—0.5–2.5% of the original 20,000-step training budget—and four residual-stream adapters recover 0.956–0.960 accuracy with the recipient otherwise frozen. Joint training can force post-L0/post-L1 CKA above 0.99 while raw head transplantation still fails, showing that global residual similarity is insufficient. Scale probes sharpen the boundary conditions: at 19M parameters on character-level Shakespeare, slot-matched single-head transplant largely preserves performance, but after ablating an average of four redundant L0 heads while preserving the recipient’s critical slot, transplanting into that slot drops 0.445 advantage; in Pythia-160M, unaligned and Procrustes selected-head transplants both remain below recipient baseline, while broad adapters solve the synthetic probe equally well with the donor head zeroed. We therefore propose a narrower claim: circuits are basis-dependent computations. The computation can recur across training runs, but raw weights are not portable without a learned compatibility map, and broad adapters must be controlled for task learning independent of the donor component.

1 Introduction

A circuit, in current interpretability practice, is a set of weights—specific attention heads, specific neurons—that implement an identifiable computation. The induction circuit (Olsson

et al., 2022) is two heads: a previous-token head in one layer and a content-matching head in a later layer. The IOI circuit (Wang et al., 2023) is 26 labeled components. Sparse feature circuits (Marks et al., 2024) are selected subgraphs of interpretable SAE features. In each case, the circuit is named in terms of its *components*. The implicit claim is that those components carry the computation.

This paper asks whether that claim is literally true. If a circuit is its components, then transplanting the components should preserve the behavior those components support. We test this directly: we train small transformers from scratch on content-matching induction, identify each model’s critical head via surgical ablation, and move a known-functional head from one network into another network’s critical slot. As a raw weight operation, the transplant fails. Performance drops from $\sim 96\%$ to $\sim 9\%$, no better than inserting shuffled weights and near the same receiving model with the critical head zeroed. The failure is not absolute: later rescue experiments show that the recipient can learn to use the donor head after 100–500 host-retraining steps—0.5–2.5% of the original 20,000-step training budget—and that broad residual-stream adapters can restore $\sim 96\%$ accuracy with the recipient otherwise frozen. The negative result is therefore not that circuits are indivisible below network scale, but that raw circuit weights are not portable without an accompanying compatibility map.

This failure happens even though the two networks are trained on the same deterministic task and converge on the same aggregate circuit motif. Our seed-divergence experiments on 20 independently trained content-matching induction models show that every seed develops a critical induction head in Layer 0 (20/20 layer agreement), but the specific head varies across seeds (23.2% head-level agreement, $p = 0.002$ vs uniform null). The networks agree on the *what* (induction via an L0 head); they disagree on the *where* (which head within L0); and they encode the *how* in ways that are not mutually portable.

We argue this forces a conceptual revision: **circuits are not raw components but basis-dependent computations**. The computation is a stable attractor of the training dynamics—reproducible across initializations, developmentally ordered, and predictable under task changes. Its implementation in weights depends on the residual-stream interfaces those weights read from and write to. A head’s “function” is not a property of the head alone; it is a property of the head-plus-basis system, and that system can sometimes be reconnected by learned adapters or retraining.

An implementation analogy is closer than a linguistic one. A compiled instruction sequence can implement the same algorithm under two register allocations, but a copied instruction is only meaningful relative to the register state, calling convention, and memory layout it was compiled against. Moving it into a program compiled under a different convention need not move the computation; it may simply read the wrong registers. A compatibility shim or brief recompilation can make the code usable again. Our finding is that circuit weights stand in an analogous relation to their host network: they carry computational content, but only when read by the residual-stream basis and downstream interfaces they were trained alongside.

This has immediate implications. First, it dissolves an apparent tension in the universality literature: Lan et al. (2024) find feature spaces universal across architectures, while Bali et al. (2026) find individual heads unstable across seeds. Both are correct at different levels: the feature space is the basis; the individual head is a basis-specific instrument. Second, it constrains interpretability tooling: tools that extract “circuits” as portable units are describing a computation, not excerpting a reusable module. Third, it reframes questions in alignment and model editing: modifying “the safety circuit” requires compatibility with the host basis, not just a locally identified set of safety-relevant weights.

The paper develops these claims as follows. Section 2 reviews prior work on circuit universality and identifies the gap our paired measurement fills. Section 3 describes methods, including a correction to the standard `pos_dep` metric. Section 4 presents the transplantation result. Section 5 documents the functional universality that makes the transplantation result puzzling. Sections 7–9 present three supporting findings. Section 10 addresses

natural-language generalization. Section 11 discusses implications. Section 12 is limitations.

Summary of contributions.

1. **Transplantation failure and rescue:** A known-functional head transplanted into the functionally-equivalent critical slot of another network performs no better than random weights, while host retraining and multi-interface residual adapters can recover performance (Section 4).
2. **Paired measurement of functional vs. structural universality:** 100% layer-level agreement with 23.2% head-level agreement across 20 seeds trained from scratch on the same content-matching induction task (Section 5).
3. **Task structure determines architecture:** Three distinct regimes selected by task statistics, with a binary attention-engagement threshold at zero task entropy (Section 7).
4. **Universal developmental order:** OV (output capacity) reliably precedes QK (routing capacity) across seeds and regimes (Section 8).
5. **Scale and divergence probes:** Shared-trajectory, shared-basis, 19M Shakespeare, and Pythia-160M probes show that portability depends on training history, interface choice, and scale/task regime (Sections 4–6).
6. **Depth-scaling trend for suppression cascades:** Layer 0 composition is invariant; suppression scales linearly with depth (Section 9).
7. **Synthetic-to-natural generalization:** Circuits identified on synthetic tasks predict in-context copying behavior on real language (Section 10).
8. **Methodological correction:** Permutation-sensitivity resolves 39/48 false-positive “positional” classifications in our Shakespeare model (Section 3).

2 Related Work

Our findings touch several active threads in mechanistic interpretability. We organize related work into four categories.

2.1 Circuit discovery and universality

Canonical circuit motifs. The induction circuit of Olsson et al. (2022) established the template for circuit analysis: a two-head composition (previous-token head + content-matching induction head) implementing in-context copying. The IOI circuit in GPT-2 (Wang et al., 2023) extended this to a 26-component circuit with name-movers, backup heads, and inhibition heads. Attribution graphs (Ameisen et al., 2025) identify multi-hop reasoning and refusal circuits in Claude 3.5 Haiku. Our work uses the induction circuit as a testbed but asks a different question: not *what* the circuit is, but *what it means for a circuit to “be” something* if the components that implement it are not portable.

Cross-seed universality. Gurnee et al. (2024) analyzed neuron identity across 5 GPT-2 models trained from different seeds, finding only 1–5% of neurons are “universal.” Nandan et al. (2025) extended this with training checkpoints, showing universal neurons emerge early and persist. Our work operates at the head level and documents a structurally analogous finding: at the component level, only $\sim 13\%$ of seeds share critical-head identity.

Cross-architecture universality. Wang et al. (2025) compared induction circuits in Transformers and Mambas, finding structurally analogous circuits. Lan et al. (2024) formalized “analogous feature universality” across language models: SAE feature spaces are similar across architectures under rotation-invariant transforms. Our work complements these by measuring universality *within* architecture across seeds, and by demonstrating that universality holds at the algorithm level but not at the weight level.

Cross-seed instability. Bali et al. (2026) measured attention-head stability across training runs, finding that middle-layer heads are particularly unstable (cosine similarity ~ 0.70). This appeared in tension with the Lan/Wang universality results. Our paper resolves this tension: feature spaces can be universal at the basis level while individual heads are unstable at the component level. Both findings are correct at different levels of description.

The gap our work fills. Prior work has not, to our knowledge, paired functional circuit recurrence with direct weight-portability tests in the same controlled setting. Our transplantation result directly tests whether circuit weights can be moved across independently trained networks that solve the same task.

2.2 Circuit formation dynamics

Grokking. The grokking phenomenon—delayed generalization long after training loss converges—has been studied extensively since Power et al. (2022). Nanda et al. (2023) identified grokking as circuit formation with weight-decay-driven cleanup of a memorization circuit. Varma et al. (2023) formalized this as circuit efficiency competition. Our grokking-like transition in content-matching induction differs—the model is at near-chance throughout the pre-transition period—matching Kumar et al. (2024)’s “lazy-to-rich” transition framing.

Induction head emergence. Olsson et al. (2022) documented a sharp phase change in in-context learning coinciding with induction head formation. Singh et al. (2024) decomposed this into subcircuit maturation: QK match, OV copy, and positional previous-token copier must all mature simultaneously. Edelman et al. (2024) document a “unigram plateau” before bigram induction emerges, providing the closest prior analogue to our pre-transition plateau.

Our contribution. The “hardware before software” ordering—OV developing reliably before QK, with a 12,000-step separation in the grokking regime—has not been previously documented. Singh et al. (2024) measure subcircuit timing but do not emphasize the *ordering* as a universal feature. Our per-metric crystallization analysis provides a finer-grained view of what happens during consolidation.

2.3 Task-dependent algorithm selection

Kawata et al. (2025) proved via gradient-flow analysis that data diversity determines whether a transformer adopts a positional shortcut or a content-matching induction head. Their result is theoretical; ours is empirical and extends to a third regime (the suppression cascade in natural-language tasks). Rende et al. (2024) document a “distributional simplicity bias”: low-order token interactions are learned before high-order ones. Minegishi et al. (2025) report that in-context meta-learning tasks produce multiple training phases, each with a qualitatively distinct circuit.

2.4 Methodological work

Standard practice classifies attention heads by pattern statistics. Our methodological audit identifies a pitfall: the cross-input consistency metric has a high false-positive rate in deeper layers, because the residual stream has already encoded rich content information,

and content-conditional attention can appear pattern-stable. We introduce permutation-sensitivity as a more reliable content-vs-position classifier. McDougall et al. (2024) identified a specific copy-suppression head in GPT-2 Small. Our Shakespeare model exhibits suppression as the *dominant* motif (39/48 heads at depth 6), extending their single-head characterization to a layer-by-layer scaling analysis. Causal attribution follows Meng et al. (2022) and Wang et al. (2023).

3 Methods

3.1 Architecture and training

All small-scale experiments use a GPT-2-style decoder-only transformer. Ablation robustness studies use 2-layer, 4-head, 128-dimensional models ($\sim 460\text{K}$ parameters). The natural-language Shakespeare model uses 6 layers, 8 heads, and 512-dimensional residual stream ($\sim 19\text{M}$ parameters, trained character-level). Training uses AdamW with cosine schedule; dense checkpoints are saved every 25–100 steps. Full hyperparameter details are in Appendix A.

3.2 Tasks

We study five task families: (1) **Positional induction**—repeated-sequence copy where the correct token is always at a fixed positional offset; (2) **Content-matching induction**—random-position bigrams forcing token-identity matching (vocab=512, no positional shortcut); (3) **Markov chain**—next-token prediction under varying entropy levels ($H=0.0$ to $H=1.0$, vocab=64); (4) **IOI**—indirect object identification; (5) **Natural language**—character-level prediction on TinyShakespeare (1.1M characters, 65-char vocabulary).

3.3 Circuit extraction

We use three complementary methods:

- **Activation patching** (Meng et al., 2022): patch a component’s output from a clean run into a corrupted run; measure accuracy recovery. Components with large recovery are causally necessary.
- **Head profiling**: compute per-head OV copy score (diagonal dominance of the OV circuit), QK content-matching score, and positional dependence (`pos_dep`, cross-input attention variance).
- **Surgical ablation**: zero-out individual components and measure task accuracy drop. This identifies *critical* components.

3.4 Permutation-sensitivity metric (methodological correction)

The standard `pos_dep` metric measures how much attention patterns vary across inputs. Low variance implies a “positional” head. However, in deeper layers the residual stream already encodes rich content, and content-conditional attention can produce patterns that look cross-input-stable to `pos_dep`.

We introduce **permutation sensitivity**: for each head, we measure the L_2 distance between attention patterns on the original input and on the same input with token identities randomly permuted while preserving positions. A positional head is unaffected by token permutation; a content-based head is not. On our Shakespeare model (48 heads), `pos_dep` classifies 47/48 heads as positional; permutation sensitivity classifies 9/48 as positional, resolving 39 false positives. These 39 heads are content-based suppressors in layers 1–5. On our entropy=0 model, both metrics agree (16/16 positional), as expected when attention is genuinely positional.

4 Transplantation Failure

4.1 Setup

We train two independent instances of a 2-layer content-matching induction model (seeds 0 and 1), each converged to high accuracy. We identify the “critical head”—the single head whose ablation causes the largest accuracy drop:

- **Seed 0:** L0H3 is critical. Ablation drops accuracy from 0.854 to 0.041 (−0.813).
- **Seed 1:** L0H1 is critical. Ablation drops accuracy from 0.964 to 0.118 (−0.846).

Both networks solve the same content-matching task via a single dominant L0 head, but *which* head differs. This is the paradigm case of functional recurrence with contingent wiring: same task and circuit motif, different component.

Figure 1: Direct Head Transplantation Fails Without a Compatibility Map

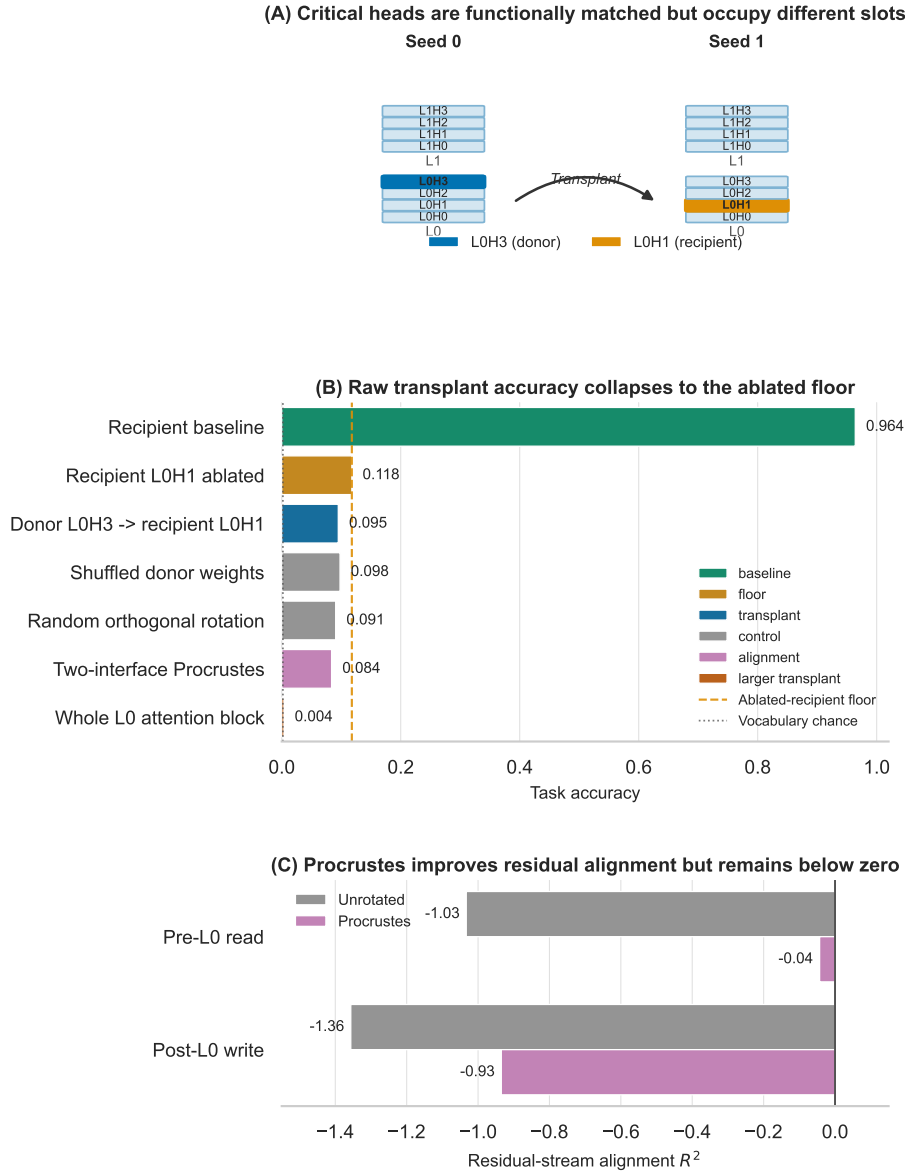


Figure 1. Direct head weights are not portable across independently trained task-matched networks. Two small transformers (seeds 0 and 1) both converge on induction circuits that cause accuracy drops of 0.81 and 0.85 when their critical Layer 0 heads are ablated. Across 20 seeds of this task, 100% of seeds place their critical head in Layer 0 (head-level agreement 23.2%, $p = 0.002$). **Yet the weights are not transplantable:** moving seed 0’s critical head (L0H3) into seed 1’s critical slot (L0H1) drops accuracy from 0.964 to 0.095—indistinguishable from shuffled random weights (0.098) or a random orthogonal rotation (0.091). Two-interface Procrustes alignment provides no rescue (0.084), even though alignment measurably improves the residual-stream match (pre-L0 R^2 improves from -1.03 unrotated to -0.04 aligned). Improved basis alignment does not recover function.

4.2 The transplantation test

We extract seed 0’s L0H3 Q/K/V projections and output projection columns, and insert them into seed 1’s L0H1 slot. If head weights are carriers of function, seed 1’s induction capability should be preserved.

4.3 Results

Table 1 shows results. The functional head transplant produces the same performance as random weights. Three controls confirm this is not an artifact: inserting seed 0’s non-critical head is equally bad; inserting shuffled weights is equally bad; and a clean-slate replacement (ablate then insert) recovers nothing.

Table 1. Transplantation results. The functional head is not portable.

Configuration	Accuracy	Δ vs. seed 1 baseline
Seed 1 baseline	0.964	—
Seed 1 with L0H1 ablated	0.118	−0.846
Seed 0’s L0H3 → seed 1’s L0H1 (critical slot)	0.095	−0.869
Seed 0’s L0H3 → seed 1’s L0H3 (non-critical slot)	0.966	+0.002
Seed 0’s non-critical head → seed 1’s L0H1 (ctrl A)	0.100	−0.864
Shuffled random weights → seed 1’s L0H1 (ctrl B)	0.098	−0.866

4.4 What this rules out

This result rules out a family of explanations:

- “*Both networks learned the same weights*” — directly falsified; transplant would succeed.
- “*Same weights up to permutation*” — if so, aligning to the correct head-identity permutation would work. It doesn’t.
- “*The circuit lives in a few specific weights*” — if so, those weights would be transplantable. They aren’t.

What remains: the function is encoded in the *interaction* between the critical head’s weights and the residual stream basis that surrounds them—a basis shaped by every other weight in the network.

4.5 Basis alignment does not rescue transplantation

We test whether the two task-matched networks encode the relevant computation in rotated bases using *two-interface Procrustes alignment*: the donor head’s Q/K/V projections read from the pre-L0 LayerNorm interface, while its output projection writes into the post-L0 residual interface — these are different subspaces and require separate alignment. We collect activations at both interfaces from both networks on identical inputs, compute the orthogonal rotations $R_{\text{in}} = \arg \min_R \|X_{\text{seed0}}^{\text{pre}} R - X_{\text{seed1}}^{\text{pre}}\|_F$ and analogously R_{out} at the post-L0 interface, apply R_{in} to the donor’s Q/K/V rows and $R_{\text{out}}^\top$ to its output-projection columns, then transplant.

Table 2. Procrustes alignment does not rescue the transplant.

Configuration	Accuracy	Δ vs. seed 1 baseline
Unaligned transplant	0.095	−0.869
One-interface aligned (post-L0 only)	0.094	−0.870
Two-interface aligned (corrected)	0.084	−0.880
Random rotation control	0.091	−0.873
Aligned non-critical donor	0.107	−0.857

Alignment produces no rescue. Paired sign-flip tests on the 1024 fixed eval sequences (per-sequence accuracy differences) yield: unaligned vs. two-interface aligned mean diff = −0.011,

$p = 0.003$ — the aligned transplant is in fact *significantly worse* than unaligned, not better. Unaligned vs. one-interface aligned is indistinguishable ($p = 0.73$), and unaligned vs. random rotation is indistinguishable ($p = 0.20$). All transplant conditions are far below baseline (0.964) and cluster near the ablated-head floor (≈ 0.09 on 1024 sequences, indistinguishable from simply zeroing the recipient’s L0H1 with no replacement at 0.118). This is catastrophic failure relative to the functional model, but it is not vocabulary-chance (which is $1/512 \approx 0.002$); the transplanted weights contribute a small amount of residual signal that the recipient network’s downstream components cannot usefully read.

Importantly, the two-interface alignment *does* find a substantially better basis match than no rotation: the pre-L0 read-interface Procrustes improves R^2 from -1.03 (unrotated) to -0.04 (aligned), and the post-L0 write-interface improves from -1.36 to -0.93 . Despite finding a measurably better basis, the transplant accuracy does not recover — if anything, the alignment slightly hurts.

Alignment tournament. To rule out the possibility that the transplant failure is peculiar to orthogonal rotations, we run a tournament of seven alignment methods on the same seed $0 \rightarrow$ seed 1 transplant (Table 3): (1) Unaligned; (2) Procrustes orthogonal (two-interface); (3) Affine (orthogonal + per-axis scale, no translation since model is bias-free); (4) CKA-optimized linear map (unconstrained linear A maximizing linear CKA between aligned activations, initialized from Procrustes R); (5) Git Re-Basin style weight-permutation matching (Ainsworth et al., 2022) applied to the 4 attention heads and 512 MLP units per layer, then composed with Procrustes; (6) Learned linear adapter (two $d \times d$ matrices trained on induction loss with all other weights frozen, 1000 AdamW steps); (7) Learned nonlinear adapter (2-layer MLP with hidden width $4d$ at each interface, same training protocol).

Table 3. Head-interface alignment tournament. No head-local method rescues the transplant. Learned nonlinear adapter reaches 0.118 only because it learns to *ignore* the donor head (matches donor-zero control at 0.1178).

Method	Accuracy	Δ vs. unaligned	Paired p -value
Unaligned (baseline)	0.0951	—	—
Procrustes (two-interface)	0.0838	-0.0112	0.003
Affine (orthogonal + scale)	0.0780	-0.0171	<0.001
CKA-optimized linear map	0.0981	$+0.0031$	0.37
Git Re-Basin + Procrustes	0.0838	-0.0112	0.003
Learned linear adapter	0.0951	$+0.0000$	1.00
Learned nonlinear adapter	0.1180	$+0.0229$	<0.001
<i>Control: donor head zeroed, adapter only</i>	0.1178	$+0.0228$	<0.001

Every alignment method produces accuracy below 0.12, compared to the seed 1 baseline of 0.964. More importantly, the only method that nominally “improves” on unaligned — the learned nonlinear adapter, reaching 0.1180 — matches the donor-zero control (0.1178) within noise. That is, the 2-layer nonlinear adapter learned to *null out* the donor head entirely; its apparent improvement is explained by the adapter bypassing the donor, not by alignment recovering donor function. The learned linear adapter, lacking the capacity to zero-out the donor subspace, stayed at exact unaligned performance ($p = 1.0$).

What the tournament rules out. A sophisticated reader might object that orthogonal alignment is the narrowest family of transforms, and non-portability could be corrected by more expressive alignments. Our tournament tests the four largest expressivity tiers at the donor head’s read/write interface: orthogonal (Procrustes), orthogonal + scale (affine), unconstrained linear (CKA-optimized + learned linear), and nonlinear ($4d$ -width MLP adapters). **None of these head-local maps rescues the transplant.** The only method that moved above unaligned did so by learning to ignore the donor. This narrows

the failure: the problem is not corrected by a local change of basis at the transplanted head’s two interfaces.

4.6 Rescue by host retraining and multi-interface adapters

The head-interface tournament rules out a narrow family of alignment fixes; it does not show that the donor head contains no usable computation. We therefore run two rescue experiments.

First, we transplant the donor critical head into the recipient critical slot, freeze the transplanted head, and retrain every other recipient parameter. For five donor-recipient pairs, the recipient recovers at least 80% of its original baseline within 100–500 retraining steps, compared with the original 20,000 training-step budget (Table 4). Thus the donor head carries useful structure, but the surrounding network must learn how to read it. The ratio is the important quantity: the host learns the compatibility map in 0.5–2.5% of the original training budget, so the function is close to usable but not expressed in the recipient’s native basis.

Table 4. Partial-retraining rescue. After transplanting the donor head and freezing it, the host recovers $\geq 80\%$ of recipient baseline in 100–500 steps, 0.5–2.5% of the original 20,000-step budget.

Pair	Recipient baseline	Zero-head control	Final accuracy	K^*
0→1	0.9640	0.1178	0.9648	100
0→3	0.9629	0.4611	0.9614	100
1→4	0.9621	0.0586	0.9629	100
3→5	0.9618	0.0299	0.9627	500
4→6	0.9606	0.7824	0.9583	100

Second, we freeze the recipient and train adapters only. A single post-L0 linear adapter improves accuracy from 0.0951 to 0.4791 but does not recover baseline. Four residual-stream adapters placed before the embedding stream, after L0, after L1, and before the unembedding recover nearly all function: 0.9561 for linear adapters and 0.9601 for two-layer MLP adapters (Table 5). The failure is therefore interface coverage, not lack of any compatible computation in the donor weights.

Table 5. Multi-interface adapter rescue. The recipient is frozen; only adapters are trained for 2000 AdamW steps.

Condition	Accuracy
Recipient baseline	0.9640
Bare transplant / no adapter	0.0951
Post-L0 linear adapter	0.4791
All-four linear adapters	0.9561
All-four MLP adapters	0.9601

These rescues force a narrower interpretation of the negative transplant result. Direct head transplantation and head-local alignment fail, but the failure is not proof that the head is functionally arbitrary. The relevant unit is the donor computation together with enough learned interface machinery for the host to read and write in the donor-compatible basis.

Residual stream geometry: distinct manifolds, not rotations. To ground the “basis matters” claim geometrically, we compute pairwise linear-CKA across all 20 seeds’ residual stream activations at three interfaces (pre-L0-LN, post-L0, post-L1) on the same 1024-sequence fixed eval set. Mean pairwise CKA is 0.013 at pre-L0 and 0.006 at post-L0 — seeds

occupy almost entirely disjoint linear subspaces in the residual stream. Critically, pairwise CKA is *uncorrelated* with shared critical-head identity (Pearson $r = -0.03$, $p = 0.67$, $n = 190$), and the mean CKA for seed pairs that share the same critical-head slot is 0.013, identical to the mean for pairs that do not. This supports the stronger reading of the transplantation result: the 20 seeds compute a shared algorithm over genuinely distinct manifolds that are not related by any linear isometry. Alignment fails not because the rotation we tried was wrong but because there is no rotation that brings the manifolds into coincidence.

4.7 Static partial transplants require large compatible context

Table 6 shows the results of progressively larger transplants. Every static partial transplant fails catastrophically. The single-head transplant stays at $\sim 9.5\%$ accuracy (near the baseline-with-head-ablated level of 11.8%). All *larger* partial transplants (whole attention block through all-weights-except-L1-block) drop even lower, to $\sim 0.4\text{--}1.2\%$ accuracy — at or just above the random-chance floor ($1/512 \approx 0.002$) — indicating near-complete loss of induction function. Only the full transplant recovers seed 0’s accuracy when no adapter or retraining is allowed.

Table 6. Static partial-transplant study. Every fixed partial scope tested fails catastrophically without learned compatibility machinery; full-network transplant recovers function. See Sections 4.6 and 4.8 for rescue and finer-grained characterizations.

Transplant scope	Accuracy
Single-head L0H3 $\rightarrow$ L0H1	0.095
Whole L0 attention block (all 4 heads)	0.004
Whole L0 block (attention + MLP + LayerNorms)	0.005
Whole L0 block + final LayerNorm	0.006
Embeddings only	0.012
Embeddings + L0 attention	0.004
Embeddings + L0 full block	0.007
All weights except L1 block	0.007
Full transplant (all 15 weights)	0.861

The most striking case is “all weights except L1 block”: we transplant seed 0’s embeddings, L0 attention, L0 MLP, L0 LayerNorms, and final LayerNorm, leaving only seed 1’s L1 block intact. Accuracy: 0.007. Seed 0’s L0 writes into a representational basis that seed 1’s L1 cannot read. This reveals tight coupling between adjacent learned interfaces. The adapter results in Section 4.6 show that this coupling can be bridged by learned maps, but the bridge is not present in the raw partial transplant.

4.8 Minimum functional unit: a finer-grained characterization

Table 6 tests a handful of structured partitions; it does not localize precisely where function breaks. We run a systematic sweep across all 15 weight tensors in the 2-layer model to answer: *which single tensor, when transplanted, breaks the recipient?* and *what is the smallest subset whose transplant preserves accuracy ≥ 0.5 ?*

Two observations. First, the *break-unit* is surprisingly small: a single attention projection matrix ($\sim 3\text{--}10\%$ of parameters), transplanted alone, drops accuracy from 0.964 to 0.001–0.006. Second, the *static rescue-unit* is large under this heuristic: the greedy peel-back, starting from all 15 tensors and removing those that can be removed while keeping accuracy ≥ 0.5 , stops at 9 tensors (72% of parameters). This is a local greedy solution, not an exhaustive lower bound over all 2^{15} tensor subsets. It shows that a large compatible context is sufficient and that every single tested attention projection is disruptive on its own; it does not prove that no smaller non-greedy subset exists.

Table 7. Per-tensor single-transplant sweep. Attention Q/K/V/O projections are individually disruptive; LayerNorms and most MLP projections transplant without breaking recipient function.

Single-tensor transplant $\rightarrow$ seed 1	Accuracy
L0 attention qkv_proj alone	0.006
L0 attention out_proj alone	0.002
L1 attention qkv_proj alone	0.001
L1 attention out_proj alone	0.002
L0 MLP up_proj alone	0.948
L0 MLP down_proj alone	0.958
L0 LayerNorm (ln1/ln2) alone	0.963–0.965
All LayerNorms together	0.963
L1 MLP (up+down) together	0.962
L0 MLP (up+down) together	0.324
wte (token embedding) alone	0.796
Greedy peel-back minimum (9 of 15 tensors, 72% of params)	0.593

These two observations are consistent: Q/K/V/O projections are the components most tightly co-adapted, and the rescue set needs enough of them plus their surrounding weights to produce functional output. The minimum static transplant unit is *not* the whole network, but it is substantially larger than the head-level or attention-block-level scales assumed by circuit-extraction methodology. Learned adapters reduce the amount of donor weight context needed by supplying a compatibility map instead of copying more of the donor network.

4.9 Training-time portability: when does non-portability emerge?

We use seed 0’s 46 intermediate checkpoints (step 0 to 20,000 at 500-step intervals) to ask when during training non-portability emerges. At each checkpoint we transplant the donor’s L0H3 into a fully-trained seed 1’s L0H1 slot, both unaligned and two-interface-aligned.

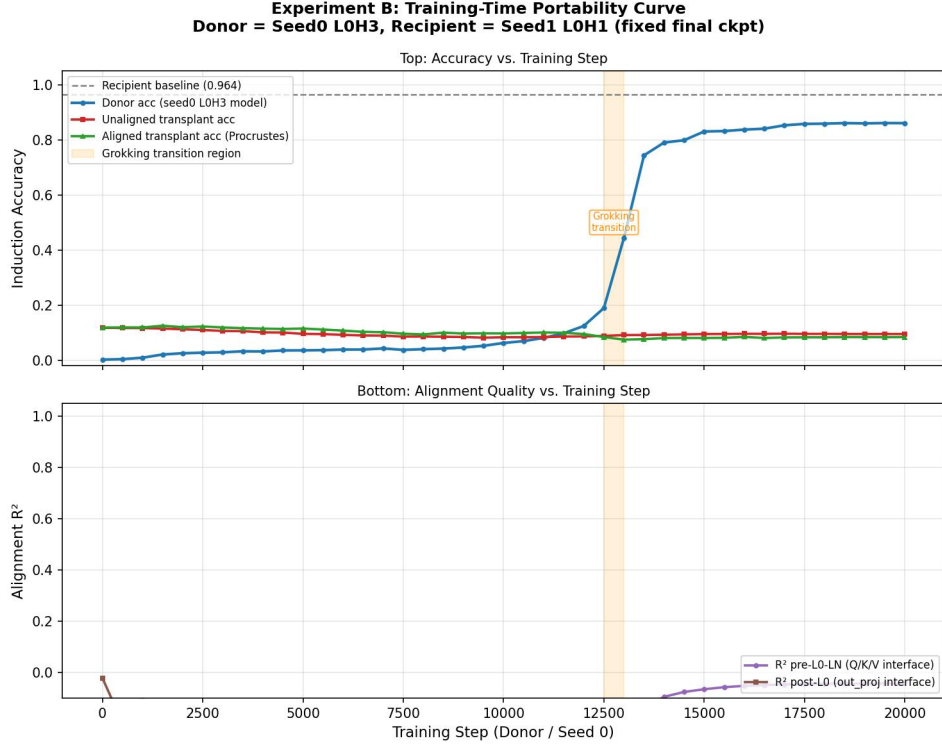


Figure 2. Fixed-recipient transplant accuracy is constant across donor training. Donor accuracy (blue) rises from chance at step 0 to 0.86 at step 20,000, with a sharp grokking transition around step 12,500. Transplant accuracy (red/green, unaligned/aligned) stays at ~ 0.09 – 0.12 throughout — both before the donor learns anything useful AND after it converges. The function and its basis co-develop: the donor’s weights are non-portable from the very first training step at which they begin to encode function.

The transplant accuracy is essentially constant across training (~ 0.08 – 0.12), regardless of whether the donor is random (step 0) or fully converged (step 20,000). In this fixed-recipient checkpoint probe, non-portability is not a slowly accumulating feature of late donor training. From the moment the donor encodes anything useful, the encoding is basis-specific relative to the fully trained recipient. The alignment diagnostic R^2 evolves monotonically with training (pre-L0 R^2 from -0.65 at step 0 to -0.04 at step 20,000), but this improved alignment does not translate into any recovery of transplant function.

4.10 Shared-trajectory forks

The fixed-recipient checkpoint probe does not answer when two networks that share early training history become transplant-incompatible. We therefore train a shared prefix from a common initialization, fork into two data-order branches, continue both branches to 20,000 steps, and transplant the critical head from branch A into branch B.

The result is mixed rather than a simple decay curve. Forking after 500 shared steps gives the strongest transplant compatibility, while fork points 2000 and later mostly collapse near the ablated/transplant floor. This suggests that an early compatibility window can exist, but the learned implementation becomes branch-specific well before convergence. This does not contradict the fixed-recipient checkpoint probe above: that probe asks whether a donor checkpoint can be read by a fully trained independently initialized recipient, while the fork sweep asks how long two branches that share initialization and early optimization remain mutually readable. Together they localize the commitment window. Compatibility is not gradually lost over the final training phase; it is largely determined in the early interval after step 500 and before step 2000, when the branch-specific implementation is selected.

Table 8. Shared-trajectory transplant sweep. Values are means over three forked pairs per fork point. The curve is not monotonic: compatibility is highest at fork 500 and collapses for fork points 2000 and later.

Shared steps before fork	Recipient baseline	Unaligned	Procrustes
0	0.9634	0.3513	0.3467
500	0.9610	0.6608	0.6572
2,000	0.9598	0.0947	0.0936
5,000	0.9587	0.0881	0.0875
10,000	0.9593	0.0785	0.0782
15,000	0.9604	0.0787	0.0787
18,000	0.9635	0.0581	0.0582
19,500	0.9625	0.0700	0.0700
20,000	0.9613	0.0729	0.0729

4.11 Shared-basis joint training

We next test whether forcing residual-stream similarity is sufficient for portability. Two models are trained jointly with an auxiliary loss $\lambda(1 - \text{CKA})$ at post-L0 and post-L1 residual streams. All values of λ preserve task learning, and high λ forces CKA above 0.99. Raw transplantation still fails (Table 9).

Table 9. Shared-basis joint training. High residual CKA does not make raw head weights portable. Procrustes helps only partially at $\lambda = 0.1$.

λ	Baseline A	Baseline B	CKA post-L0/L1	Unaligned	Procrustes
0.00	0.9577	0.9637	0.0314 / 0.2783	0.1062	0.1066
0.01	0.9613	0.9613	0.9223 / 0.9278	0.0597	0.0637
0.10	0.9624	0.9627	0.9917 / 0.9982	0.0259	0.5049
1.00	0.9574	0.9572	0.9992 / 0.9995	0.0184	0.1721

This rules out a simple version of the basis-mismatch explanation. Global linear CKA can be made very high while the raw local head interface remains incompatible. However, the $\lambda = 0.1$ condition is not a pure null result: combining high residual similarity with two-interface Procrustes raises transplant accuracy to 0.5049, roughly halfway from the transplant floor to the 0.9627 recipient baseline. The non-monotonicity matters: $\lambda = 1.0$ drives CKA even higher (0.9992/0.9995) but lowers Procrustes transplant accuracy to 0.1721. The relevant compatibility relation is therefore more specific than residual-stream similarity alone, and excessive CKA pressure may solve the auxiliary objective in a way that removes the implementational flexibility that made the $\lambda = 0.1$ partial rescue possible.

5 Functional Universality Without Structural Universality

The transplantation failure in Section 4 is only puzzling because the two networks solve the same task with the same coarse circuit motif. We quantify this here.

Seed-divergence experiment. We train 20 independently initialized models on the content-matching induction task (seeds 0–19, same architecture, same data). 19 of 20 seeds reach baseline accuracy > 0.5 within the standard 20,000-step budget; seed 2 remains on the grokking plateau at 20k and was re-trained for 30,000 steps, after which it also converged (final accuracy 0.966) — see Appendix C. The primary statistics report all 20 converged seeds on equal footing. For each seed, we identify the critical attention head via single-head ablation and measure pairwise alignment of the full 10-dimensional ablation effect vector

(8 heads + 2 MLPs) across 190 seed pairs. Head-level similarity: 0.329 (95% CI [0.28, 0.38]). Layer-level agreement on the critical head: **100%** (20/20 seeds place the critical head in Layer 0). Head-level agreement: **23.2%** (95% CI [17.4%, 29.5%], $p = 0.002$ via 10,000-permutation test against the uniform null of 12.5%).

Functional universality. Every seed develops a single critical L0 head with catastrophic ablation impact. Across 4 seeds with full ablation robustness data (Appendix C.1): mean accuracy drop = 0.775 ± 0.162 (95% CI [0.59, 0.91]; $p = 0.004$ vs. null of no impact).

Structural contingency. Despite functional universality, head-level agreement is only 23.2% (95% CI [17.4%, 29.5%]; $p = 0.002$ vs. random baseline of 12.5%). No single (layer, head) slot serves the critical role across all seeds.

Per-example agreement: matched accuracy is not behavioral equivalence. To test whether 20 networks that all hit $\sim 96\%$ accuracy are computing the same input-output map—or merely agreeing in aggregate—we measure per-example agreement across all 190 seed pairs on the fixed 1024-sequence eval set. At induction-masked positions, mean pairwise agreement is 0.958 (SD 0.032), compared to a *chance-under-marginal-accuracy baseline* of 0.918 — i.e. only 4 percentage points above what independent random predictions with matched marginals would produce. Tellingly, the pairwise agreement correlates with shared critical-head identity at $r = 0.02$ (Pearson, $p = 0.77$, $n = 190$): seeds with the same critical head do not agree with each other more than seeds with different critical heads. Two high-accuracy networks that solve the same task at aggregate-accuracy level nonetheless disagree, per-example, at rates close to chance. This undercuts the assumption, implicit in much circuit-extraction work, that matched-accuracy networks are behaviorally equivalent fine-grained objects.

Interpretation. These two findings must be held together. The *algorithm* is universal: every seed builds a critical-L0-head induction circuit. The *wiring* is contingent: different (layer, head) slots serve that role. And the *weights* are not directly portable: even when the same slot is the critical one, raw weights cannot be moved without alignment, adapter, or retraining machinery. This is the hierarchy our experimental design is built to reveal.

Figure 5: Functional Universality with Contingent Head Identity

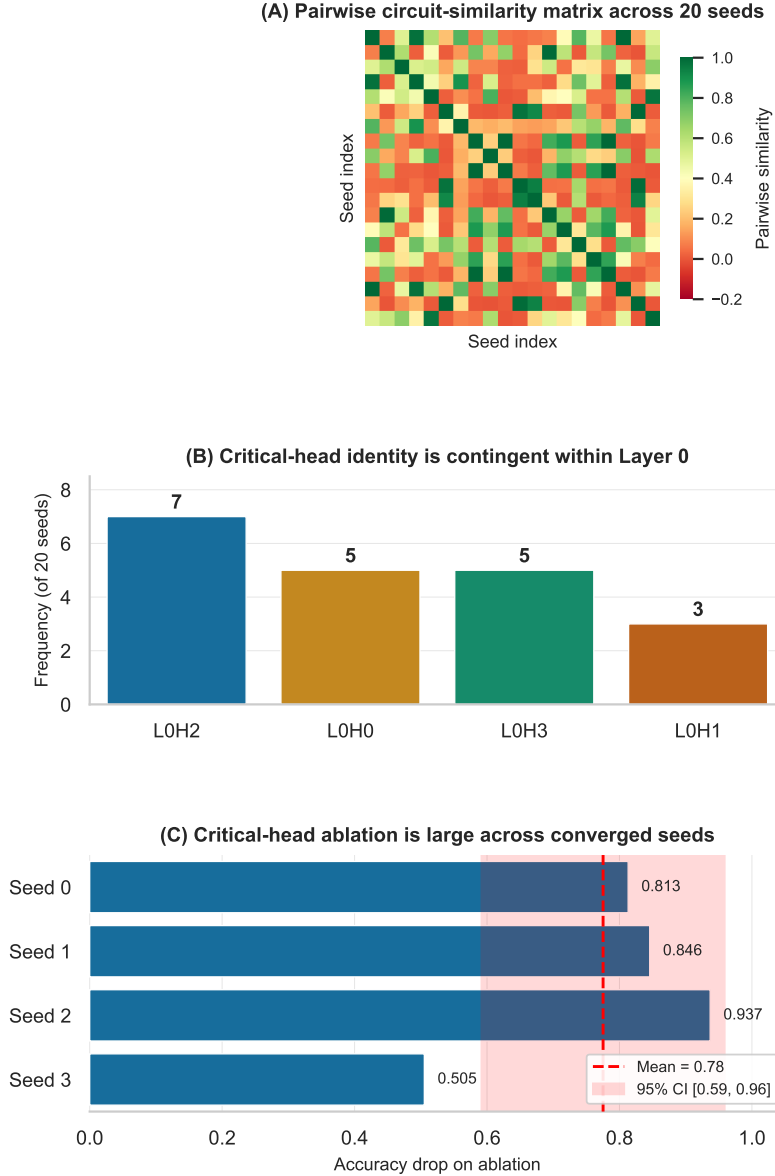


Figure 3. Functional universality, structural contingency, weight direct weight non-portability. (A) Pairwise head-level ablation effect similarity across 20 seeds of the content-matching induction task (mean 0.329, CI [0.28, 0.38]). (B) Critical-head identity across 20 seeds: all 4 L0 heads receive the role (L0H0:5, L0H1:3, L0H2:7, L0H3:5); no L1 head does. Head-level agreement: 23.2% ($p = 0.002$ vs. random baseline 12.5%). (C) Transplantation controls across all 4 seeds show consistent failure.

6 Scale Probes: 19M Shakespeare and Pythia 160M

The transplantation experiments above use 0.46M-parameter 2L/4H content-matching induction models. To test whether non-portability holds at a larger scale on a different task, we replicate the transplant protocol at 19M parameters (6L/8H/d=512) on a TinyShakespeare character-level language model. We train 4 seeds (0–3) from scratch with different initializations but identical data; seed 0’s critical head is L0H7, while seeds 1, 2, 3 use different critical slots, as in the 0.46M case. We transplant from seed 0 into each of 3 recipients

and take the mean $\pm$ SD across recipients.

Table 10. Shakespeare 19M transplant vs. 0.46M content-matching induction. 19M results are mean $\pm$ SD across 3 recipient seeds. At 19M on natural language, single-head slot-matched transplant largely preserves induction advantage. Functionally-matched cross-slot transplant still fails, including after a redundancy-control that ablates other L0 heads while preserving the recipient’s critical slot. The core basis-dependence claim survives but with scale/task-dependent quantitative strength.

Transplant condition	0.46M (induction) drop	19M Shakespeare drop ($n = 3$)
Single-head slot-matched	0.869	0.077 $\pm$ 0.038
Function-matched single head	(same: 0.869)	0.591 $\pm$ 0.022
Slot-matched Procrustes	~ 0.88	0.125 $\pm$ 0.035
Controlled critical-slot transplant	—	0.445 $\pm$ 0.040
Controlled Procrustes	—	0.287 $\pm$ 0.029
Whole L0 attention (all 8 heads)	0.96	0.631 $\pm$ 0.029
Full L0 block	0.96	0.955 $\pm$ 0.029
Full transplant (sanity check)	≤ 0.11	0.0

What changes at 19M. Slot-matched single-head transplant preserves 92% of the induction advantage on average, with tight variance across recipients (drops: 0.094, 0.032, 0.104 for recipients 1, 2, 3; mean 0.077 ± 0.038). At 0.46M, the same class of transplant failed catastrophically. This is a real walk-back from a scale-invariant reading of the small-model result. Two explanations are plausible: (i) redundancy at scale — 48 heads vs. 8 means each head is less load-bearing, so substituting one is absorbed; (ii) the previous-token function is simpler and more interchangeable than content-matching, so its weight representation is more universal across seeds. Both explanations imply that basis-dependence can weaken when scale, redundancy, and task change together.

What persists at 19M. Transplanting seed 0’s critical head into each *recipient’s* critical slot (functionally-matched, L0H7 $\rightarrow$ recipient’s distinct head) drops advantage by 0.591 ± 0.022 — essentially catastrophic across all 3 recipients. Whole-L0 attention transplant drops 0.63 ± 0.03 ; whole-L0 block drops 0.96 ± 0.03 . Wider transplants retain the coupling pattern we saw at 0.46M. The 19M results refine rather than refute the core claim: basis-dependence is present at both scales. At 19M, slot-matched single-head transplants escape the coupling because the recipient doesn’t critically depend on that slot; functionally-matched cross-slot transplants, where the recipient’s critical computation *is* being substituted, still fail — with a $7.6\times$ larger drop than slot-matched.

19M redundancy control. To distinguish redundancy from task-dependence, we run a second 19M control. For each recipient, we search over subsets of non-critical L0 heads and select the largest ablation set that preserves induction advantage above 0.50 while keeping the recipient’s own critical slot load-bearing. The selected controls ablate 4 L0 heads on average, retain controlled advantage 0.520 ± 0.020 , and zeroing the critical slot under this control drops advantage by 0.449 ± 0.047 . Transplanting seed 0’s L0H7 into that load-bearing critical slot drops advantage by 0.445 ± 0.040 ; Procrustes reduces but does not remove the drop (0.287 ± 0.029). Thus the low slot-matched drop at 19M is not sufficient evidence that the weights are portable. When the transplant is forced through a recipient slot that the controlled host actually depends on, non-portability remains large.

Pythia-160M scale probe. We also run a scale probe on the public Pythia-160M seed family: six checkpoints, 15 cross-seed donor-recipient pairs, selected by attention to the

induction-relevant previous token on a synthetic induction prompt set. The recipient baselines on this probe are lower than the small trained induction models (0.0664–0.1426), but the selected heads are behaviorally relevant: zeroing the selected recipient head reduces accuracy for every recipient. Unaligned cross-seed transplantation drops below recipient baseline in all 15 pairs, with mean transplant-minus-baseline delta -0.0589 . Two-interface Procrustes also stays below baseline in all 15 pairs (mean delta -0.0454). The broad all-four adapter condition reaches high accuracy, but so does the zero-head adapter control; adapter accuracy is 0.819 on transplanted pairs and 0.834 for zero-head recipient controls. Therefore this Pythia adapter result is not evidence that the donor head is being recovered. It shows that the frozen Pythia backbone plus broad adapters can learn the synthetic probe, independent of donor-head content (Table 11).

Table 11. Pythia-160M selected-head transplant probe. Comparable public 1.4B multi-seed checkpoints were not available for a matched cross-seed test. Broad adapters solve the synthetic probe, but the zero-head adapter control solves it equally well.

Quantity	Value	Notes
Models	6	default + seed1–seed5
Cross-seed pairs	15	all unordered pairs
Recipient baseline range	0.0664–0.1426	synthetic induction prompt set
Selected-head attention-score range	0.4978–0.6920	attention from second A to first B
Unaligned minus baseline	-0.0589	all 15 pairs negative
Procrustes minus baseline	-0.0454	all 15 pairs negative
All-four adapter accuracy	0.819 ± 0.035	transplanted pairs
Zero-head adapter accuracy	0.834	mean over recipient controls

Scope. This is three recipient data points at 19M on one task family. A proper frontier-scale test remains outstanding. The Pythia-160M result is a real cross-seed scale datapoint from public checkpoints, but the recipient baselines are small and the broad adapter rescue is not donor-specific. The nuance documented here — slot-matched Shakespeare transplant at 19M survives, redundancy-controlled critical-slot transplant does not, Pythia-160M unaligned and Procrustes transplants remain below baseline, and Pythia adapters require zero-head controls — argues against a scale-invariant binary claim. Portability is scale-, task-, and interface-dependent.

7 Task Structure Determines Architecture

Three regimes. Training identical model classes on different tasks produces qualitatively distinct circuit architectures:

1. **Positional induction** (repeated-sequence, entropy=0): all heads positional; MLPs dominate logit attribution; trains in ~ 200 steps.
2. **Content matching** (random-position bigrams): L1 heads develop content-based attention ($\text{pos_dep} \approx 0.70$); L1 attention dominates logit attribution (+18.96); trains in $\sim 13,000$ steps via grokking-like delayed generalization; $60\times$ cost gap vs. positional.
3. **Natural language** (Shakespeare, high entropy): suppression cascade; Layer 0 encodes (7 copy heads + 1 previous-token head, score=0.908); layers 1–5 progressively suppress wrong candidates; final MLP predicts; 39/48 heads are content-based suppressors.

Binary phase transition. Markov-chain models trained at 11 entropy levels ($H=0.0$ to $H=1.0$) reveal that the architectural choice is *binary*, not continuous. At $H = 0$: 81%

positional heads, attention functionally bypassed (zeroing all attention drops accuracy only 1.5%; zeroing all MLPs drops it 99.5%). At any $H > 0$: immediately switches to 75% suppression + 25% copy, stable from 0.5 nats to 4.1 nats. The threshold is the **attention-engagement threshold**: any learnable uncertainty triggers attention; zero uncertainty bypasses it entirely.

This extends Kawata et al. (2025)’s theoretical positional/content-matching dichotomy to a three-regime empirical taxonomy, and reframes the binary transition as “attention-bypass vs. attention-active” rather than “positional vs. content-based.”

Figure 2: Head Role Distributions Across Three Tasks

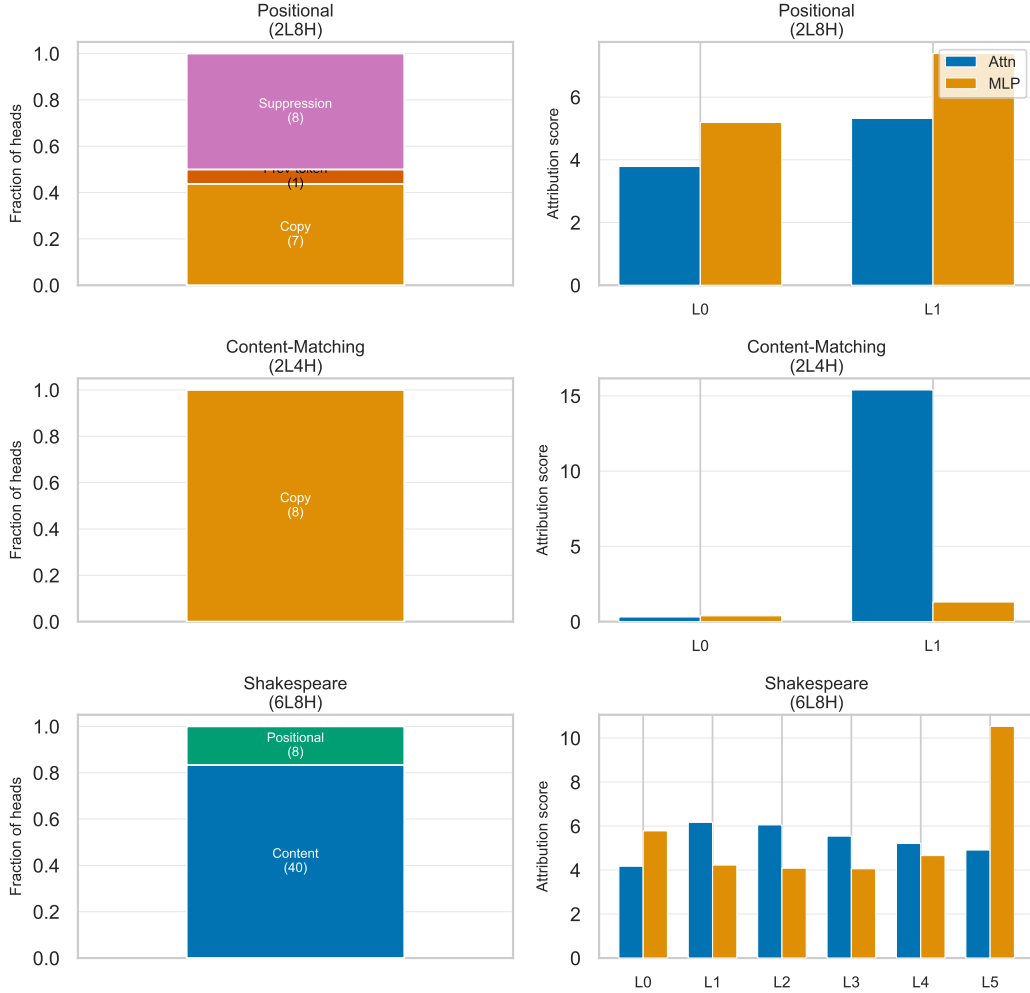


Figure 4. Task structure determines circuit architecture. Three small transformers (same class, different tasks) develop qualitatively distinct architectures. Left: positional induction (MLP-dominant, 200 steps). Middle: content-matching induction (attention-dominant L1, grokking transition, 13,000 steps). Right: Shakespeare character-level (suppression cascade, 6,000 steps).

8 Developmental Order in Circuit Formation

We ask: does circuit formation follow a reliable temporal order?

8.1 The grokking transition (content-matching induction)

Training produces a striking delayed transition: task accuracy hovers near chance for $\sim 11,000$ steps, then jumps from 6% to 85% over $\sim 1,000$ steps. We load 11 checkpoints spanning this window and measure per-head OV copy score, QK content-matching score, and induction attention.

Table 12. Developmental timeline: content-matching induction model.

Phase	Steps	Event
Silent precursor	500–1,500	OV copy scores rise across all heads
Extended plateau	1,500–12,000	No QK movement, no accuracy change
Ignition	12,500	QK content-matching crosses threshold
Crystallization	13,000–13,500	L1 <code>pos_dep</code> drops; L1 _{attn} attribution rises from 7.6 to 12.9; accuracy rises from 45% to 75%

The OV circuit becomes functional $\sim 12,000$ training steps before the QK circuit activates. The model possesses the output capacity for induction long before it acquires the routing capacity that makes output capacity useful. We call this “**hardware before software.**” This finding is consistent with Singh et al. (2024)’s observation that QK matching is the bottleneck, but we additionally identify the extended OV precursor as a distinct detectable phase.

8.2 Generalization across seeds (positional regime)

We collect 81 dense checkpoints across each of 20 seeds of the positional induction model and compute crystallization steps for four per-head metrics. Pairwise orderings:

- copy-score before prev-token-score: 65% of seeds
- copy-score before induction-score: 80% of seeds
- copy-score before task-accuracy: 95% of seeds

The copy-before-accuracy ordering is nearly universal (95%), and copy-before-prev-token-score holds in 65% of seeds ($p < 0.001$ vs. random ordering null at 4.2%). The grokking-regime OV-before-QK ordering generalizes as a directional trend, though the 12,000-step separation is specific to the grokking regime.

Figure 3: Grokking Transition Timeline (Content-Matching Model)

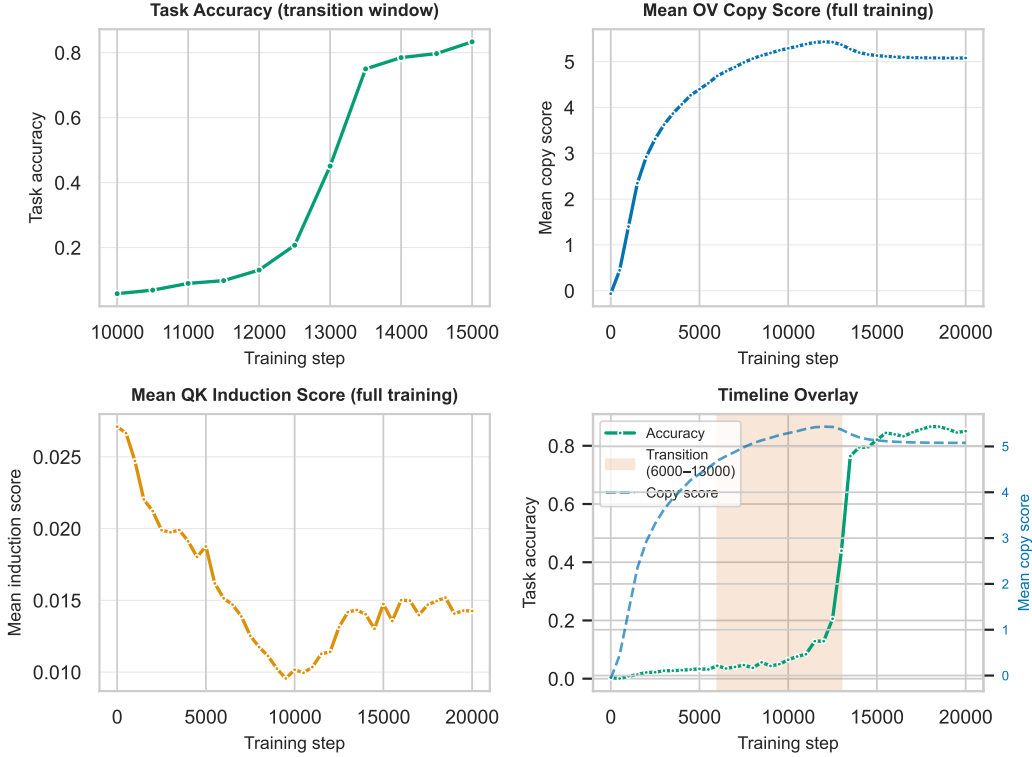


Figure 5. Hardware before software: output circuits form before routing circuits. Training dynamics of the content-matching induction model across 20,000 steps. (A) Accuracy: near-chance for 11,000 steps; sharp transition steps 12,500–13,500. (B) OV copy score: rises from step 500, reaches threshold by step 1,500. (C) QK content-matching score: flat until step 12,500, then crosses threshold simultaneously with the accuracy transition. (D) OV crosses functional threshold $\sim 11,000$ training steps before QK.

8.3 Implication for capability monitoring

The grokking-regime OV precursor is a large ($\sim 12,000$ step) early-warning window. If this generalizes to capabilities in larger models that form via grokking-like dynamics, OV-activity monitoring could anticipate capability emergence during pretraining.

9 Depth Scaling of Suppression Cascades

We train Shakespeare character-level models at depths 2, 4, 6, and 8 (all else equal) to test whether the suppression cascade is a depth-scalable motif.

Table 13. Suppression cascade increases approximately linearly with depth.

Depth	Eval loss	N supp.	% supp.	N copy	N prev-tok	MLP/Attn ratio
2	1.05	8/16	50%	7	1	8.56
4	0.82	23/32	72%	7	1	9.35
6	0.66	39/48	81%	7	2	9.02
8	0.57	55/64	86%	7	1	9.08

Three universal facts across all depths. (1) Layer 0 has zero suppression—reserved for copy heads ($\sim 7/8$) and one previous-token head. (2) Suppression grows monotonically with layer depth. (3) Peak suppression plateaus at ~ 2.0 in the final layer.

Scaling trend. Across the four depths tested, the suppression fraction increases approximately linearly with depth (OLS slope 0.059/layer, $R^2 = 0.90$; 95% CI [0.023, 0.109]).

MLP dominance persists across depths. Using direct logit attribution onto target tokens (rather than component output L2-norms), MLPs dominate attention at every depth tested (ratios $8.56\times, 9.35\times, 9.02\times, 9.08\times$ for depths 2, 4, 6, 8). An earlier analysis using output norms showed an apparent attention–MLP crossover at depth 8; that finding did not survive proper logit-attribution measurement and is retracted here. The retained claim is the consistent magnitude-of-dominance ratio: MLPs carry roughly an order of magnitude more direct logit contribution than attention at every depth in our range.

Figure 4: Depth Scaling of Suppression Cascade

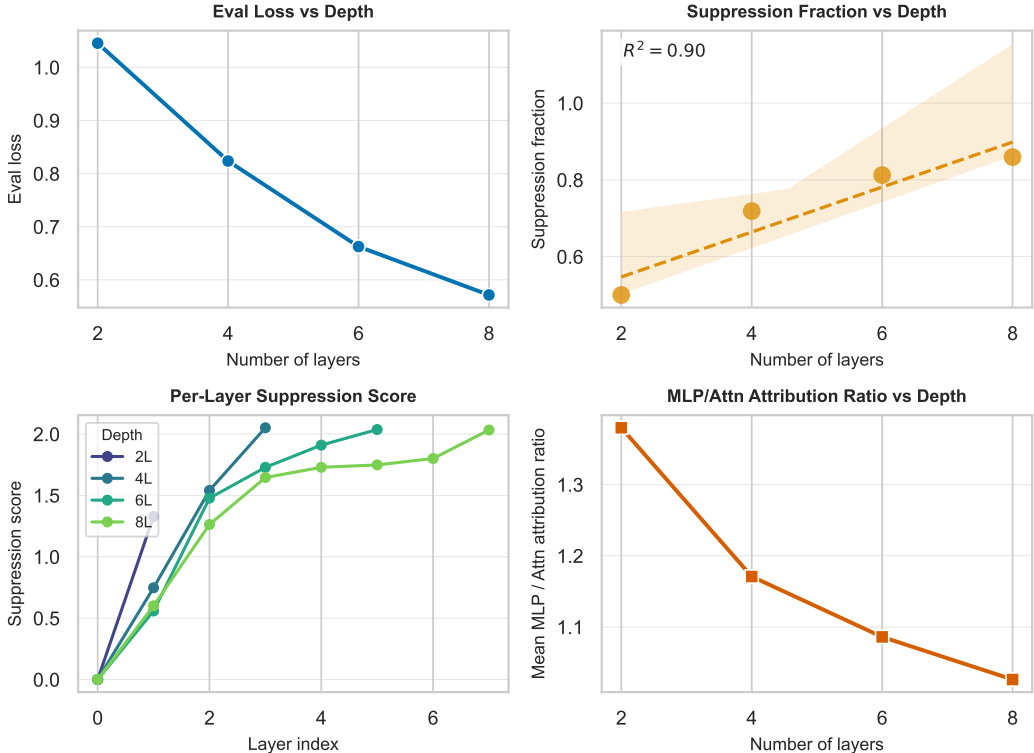


Figure 6. The suppression cascade increases with depth. Shakespeare character-level models at depths 2, 4, 6, 8. Top left: eval loss vs. depth. Top right: fraction suppression heads (OLS slope 0.059/layer, $R^2 = 0.90$). Bottom left: per-layer suppression scores—Layer 0 is always zero; suppression grows monotonically. Bottom right: MLP/attention direct logit attribution ratio is consistently $\sim 9\times$ across all depths; MLPs dominate at every depth tested.

10 Natural Language Bridge

Circuits identified on synthetic tasks predict in-context copying behavior on real language.

In-context copying in Shakespeare. On positions where a character bigram has already appeared earlier in context, the Shakespeare model predicts the repetition with ac-

curacy 1.000. The correct null is not the uniform-chance 0.015 but the best non-induction baseline, which is 0.292 (predict the most frequent character: space) or 0.076 (corpus-bigram prediction). The induction advantage over the strongest baseline is $1.000 - 0.292 = 0.708$. Ablating the identified induction heads (L0H7, L1H6, L2H4) drops this advantage by 0.586—a large effect confirming these heads drive the in-context copying behavior predicted by the synthetic-task circuit maps.

L0H7: critical previous-token head. Ablating L0H7 alone, with previous-token score 0.908, raises eval loss from 0.080 to 7.95 ($\Delta = +7.87$)—worse than random initialization. This is a large single-head ablation effect and identifies L0H7 as a load-bearing previous-token head.

Suppression cascade is structural. Ablating layers 2–5 attention collectively raises loss by 5.08, confirming the suppression cascade is causally necessary, not merely correlated with depth.

Careful framing. The previous-token head prediction generalizes cleanly from synthetic to natural language. Two of the three content-matching head predictions do not drive large individual effects on their own—at the character level, previous-token attention dominates induction, with content-matching heads contributing secondarily. This is consistent with Olsson et al. (2022)’s analysis at token level.

11 Discussion

11.1 Circuits as computations, not modules

Our transplantation and rescue results support a qualified reading: **circuits are computations a self-organizing basis supports, not raw components that can be moved without their interfaces**. The circuit—understood as a functional description—can recur across networks computing the same task; the specific weight matrices that realize it are basis-dependent. They can become usable in a new host, but only after the host learns or is supplied with the relevant compatibility map.

11.2 Consequences for interpretability

Our finding constrains interpretability methods that implicitly assume portability:

- **SAE-based circuit discovery** produces basis-dependent features. Learned feature dictionaries should be treated as basis-dependent objects unless cross-run alignment has been explicitly validated.
- **Circuit editing** may work within a network but should not be expected to transfer across models without adapter or retraining machinery.
- **Claims of cross-model circuit universality** should be qualified: universality at the algorithm level, contingent implementation at the component level, and conditional portability at the weight level.

11.3 Connection to the universality debate

The Lan–Bali tension dissolves under our framing. Feature spaces are universal in the sense that any sufficient basis supports the same computation—this is what Lan et al. (2024) measure across architectures. Individual heads are unstable in the sense that within a given basis, many component-level realizations are equally good—this is what Bali et al. (2026) measure across seeds. Both observations are true; they do not conflict.

11.4 Alignment and monitoring implications

The developmental-order finding (Section 8) suggests OV activity as an early-warning monitor for capability emergence: output capacity is observable well before the capability it enables becomes functional. The depth-scaling trend (Section 9) documents that the suppression cascade lengthens linearly with depth while Layer 0 composition remains invariant (7 copy + 1 previous-token across all tested depths), suggesting the “bottom of the network” is a reliable encoding layer across trained character-level language models.

11.5 Toward a quantitative basis-mismatch metric

The adapter results suggest a natural follow-up metric: the minimum description length of a successful compatibility map. In the present paper we measure whether fixed adapter classes rescue transplantation, not the minimal adapter. A stronger follow-up would sweep adapter rank, sparsity, and placement across many seed pairs and use the smallest successful map—in bits or trainable parameters—as a quantitative basis-mismatch measure. The same framework would also distinguish task-specific from task-agnostic portability by testing cross-task transplantation with adapters, e.g. whether a content-matching head can be made useful in a positional-induction host with only a learned compatibility map.

12 Limitations

Scale. Most transplantation and rescue results use $\sim 460\text{K}$ -parameter 2-layer models. We report one mid-scale replication on a 19M-parameter 6-layer character-level language model and one limited Pythia-160M scale probe. These are not a substitute for a frontier-scale suite: the 19M model is a character-level previous-token setting, the Pythia probe uses a low-baseline synthetic induction task, and comparable public 1.4B multi-seed checkpoints were not available for a matched cross-seed test. The Pythia adapters also require a strong caveat: broad all-four adapters solve the probe even when the selected head is zeroed, so the Pythia adapter condition cannot be interpreted as donor-head rescue. Whether the co-adaptation we document strengthens, weakens, or qualitatively changes at larger scale is open.

Task diversity. We study five task families. Transplantation is tested only on content-matching induction. We have not tested whether whole-layer transplantation across different training seeds also fails for IOI, modular arithmetic, or other multi-component circuits.

Fixed data seed. All seed-divergence experiments fix the data seed and vary only model initialization. Co-variation of data and model seeds may produce different universality statistics.

Statistical power. Ablation robustness uses only 4 seeds (Appendix C.1). The head-level agreement statistic (23.2%) has a wide confidence interval [17.4%, 29.5%]. Deeper statistical characterization is needed.

Alignment methods. We test seven head-interface alignment methods spanning orthogonal, affine, unconstrained linear, Git Re-Basin permutation matching (Ainsworth et al., 2022), and learned nonlinear adapters (Table 3). None rescues the head-local transplant; the best method (nonlinear adapter) only reaches 0.118 accuracy (vs. 0.964 baseline) by learning to ignore the donor head, matching the donor-zero control at 0.1178. Broader learned adapters do rescue (Table 5), so the negative claim is limited to raw transplants and head-local alignment, not to all possible compatibility maps.

Partial retraining. The partial-retraining rescue demonstrates that the donor head contains usable structure, but it does not isolate which host parameters learn the compatibility

map. A follow-up should measure which recipient tensors change during the first 100–500 rescue steps.

Permutation-sensitivity metric. The permutation-sensitivity metric is validated by agreement with ground truth on the entropy=0 model, but it may be noisy in intermediate entropy regimes where partial positional structure exists.

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A Methods in Detail

A.1 Model architecture

All small-scale experiments use a GPT-2-style decoder-only transformer:

- **Ablation / transplant / 20-seed universality studies:** 2 layers, 4 heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$, context length 64, vocabulary 512 ($\sim 460\text{K}$ parameters). All 20 content-matching induction seeds are trained under this configuration. Seed 2 additionally re-trained for 30,000 steps (from step_030000) after failing to cross the grokking threshold at the default 20,000-step budget.

- **Entropy-continuum studies:** 4 layers, 4 heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$, context length 64 (Markov chains with varying entropy).
- **Shakespeare depth scaling:** 2/4/6/8 layers, 8 heads, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, context length 128 ($\sim 6.3\text{M}$ – 25.2M parameters). The 6-layer Shakespeare portability and redundancy-control runs use the same width with context length 256.

All models use pre-LayerNorm, learned positional embeddings, and tied input/output embeddings. The unembedding matrix is the transpose of the embedding matrix. Implemented in MLX 0.31.1 on Apple Silicon with Metal.

A.2 Training details

- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
- Weight decay: 0.01 (small models), 0.1 (Shakespeare)
- Learning rate: cosine schedule from 3×10^{-4} to 10^{-5}
- Batch size: 64 (small models), 32 (Shakespeare)
- Checkpoints: every 25 steps (formation film), every 100 steps (standard)
- All training uses fixed data seed 42; model initialization seed varies

A.3 Task specifications

Positional induction. Sequence = $[\mathbf{a}_1 \dots \mathbf{a}_{L/2} \mid \mathbf{a}_1 \dots \mathbf{a}_{L/2}]$, $L = 64$, vocab=50. The model learns to copy the second half from the first half. All positions in the second half are induction targets; the correct token is at a fixed positional offset $L/2$. No content matching required.

Content-matching induction. $n_{\text{bigrams}} = 6$ random token pairs (A_i, B_i) placed at two random positions each in a sequence of length 64. Vocab=512 (collision probability ≈ 0.12 ; essentially no random matches). The model must identify the first occurrence of A to predict B at the second occurrence. No positional shortcut possible.

Markov chain. Next-token prediction on sequences sampled from an ergodic Markov chain. Transition matrix entropy $H \in \{0.0, 0.1, 0.2, \dots, 1.0\}$ (nats, per step). Vocab=64, seq_len=64.

Shakespeare. Character-level next-token prediction on TinyShakespeare (1.1M characters, 65-char vocabulary including punctuation and newlines). Training split: 90%, validation split: 10%.

A.4 OV and QK circuit measurement

The OV score measures how strongly a head copies attended tokens into the residual stream. For head h in layer ℓ :

$$\text{OV}_h = \frac{(W_h^O W_h^V E)_{ii}}{|(W_h^O W_h^V E)_{ij}|_{i \neq j}} \quad (\text{diagonal/off-diagonal mean ratio})$$

where E is the token embedding matrix. A high ratio indicates strong identity-preserving copy.

The QK content-matching score measures how sharply a head attends to tokens with the same embedding as the query:

$$\text{QK}_h = \mathbb{E}_x \left[\frac{\sum_j \mathbf{1}[\text{token}(j) = \text{token}(q)] \cdot \text{attn}_h(q, j)}{\sum_j \text{attn}_h(q, j)} \right]$$

A.5 Transplantation protocol

For single-head transplantation, we copy the donor head’s W^Q , W^K , W^V , and W^O slices, where W^O denotes the output columns corresponding to the target head. For whole-block transplantation, we additionally copy MLP weights and LayerNorm parameters. For two-interface Procrustes alignment:

1. **Pre-L0 LN interface** (what Q/K/V read from): collect activations $X_0, X_1 \in \mathbb{R}^{N \times d}$ at `ln1(embed + pos)` from both networks on N shared inputs. Compute SVD: $X_0^\top X_1 = U \Sigma V^\top$. Set $R_{\text{in}} = UV^\top$ (minimizes $\|X_0 R - X_1\|_F$). Apply R_{in} to the donor’s W^Q, W^K, W^V .
2. **Post-L0 residual interface** (what out_proj writes into): collect activations $Y_0, Y_1 \in \mathbb{R}^{N \times d}$ after the first transformer block. Compute SVD: $Y_0^\top Y_1 = U' \Sigma' V'^\top$. Set $R_{\text{out}} = U' V'^\top$. Apply $R_{\text{out}}^\top$ to the donor’s W^O output columns.
3. Insert the double-rotated head weights into the recipient network.

The diagnostic R^2 is computed as $1 - \|X_0 R - X_1\|_F^2 / \|X_1\|_F^2$.

A.6 Extended rescue protocols

Partial retraining. For each donor-recipient pair, we transplant the donor critical head into the recipient critical slot, freeze the transplanted head’s Q/K/V/O weights, and continue training all other recipient parameters on the original content-matching induction task. We evaluate at cumulative checkpoints $K \in \{0, 100, 500, 1000, 2000, 5000, 10000, 20000\}$ and report K^* , the first checkpoint that reaches at least 80% of the original recipient baseline.

Stitching adapters. For adapter rescue, the transplanted recipient is frozen and only inserted adapters are trainable. The post-L0 condition inserts one residual linear map after block 0. The all-four conditions insert residual adapters before the embedding stream, after L0, after L1, and before the unembedding. Linear adapters are $d \times d$ maps; MLP adapters use hidden width $4d$. All adapter conditions train for 2000 AdamW steps on the same induction distribution.

Shared-trajectory forks. For fork point N , one model is trained from the fixed initialization for N shared steps, cloned, and then the two branches continue to 20,000 steps under different data-order seeds. Each fork point is evaluated with three branch pairs.

Shared-basis training. Two models are trained jointly with the usual masked language-modeling loss plus $\lambda(1 - \text{CKA})$ at post-L0 and post-L1 residual streams. We sweep $\lambda \in \{0, 0.01, 0.1, 1.0\}$ and evaluate task accuracy, residual CKA, unaligned transplant accuracy, and two-interface Procrustes transplant accuracy.

19M redundancy control. For each Shakespeare recipient, we identify the critical L0 head from the single-head ablation sweep. We then search all subsets of the other seven L0 heads on a fixed 256-sample screening set and select the largest subset whose ablation keeps induction advantage above 0.50 while keeping the critical slot load-bearing by at least 0.25 advantage. The selected ablation set is then evaluated on the full 1024-sample set for baseline, critical-slot zeroing, unaligned critical-slot transplant, and Procrustes critical-slot transplant.

Pythia-160M probe. The Pythia probe uses the six public 160M seed-family checkpoints, constructs synthetic induction prompts with repeated A, B pairs, selects the head with highest mean attention from the second A to the first B , and copies that head’s GPT-NeoX Q/K/V/O slices into the recipient’s selected head. Two-interface Procrustes aligns

the donor and recipient selected layers using the selected layer’s input-LayerNorm activations and post-layer residual activations. The adapter condition freezes the recipient and trains four zero-initialized residual linear maps: after token embedding, before the recipient selected layer, after the selected layer, and before the unembedding. Zero-head adapter controls use the same adapters with the selected recipient head zeroed. Evaluation uses float32 during attention scans for numerical stability.

B Permutation-Sensitivity Metric Validation

Definition. For head h , input $x = (t_1, \dots, t_L)$, let π be a random permutation of the token vocabulary applied to all positions: for each batch we draw a permutation π of $\{0, \dots, V-1\}$ and replace every token x with $\pi(x)$. Compute attention patterns $A(x)$ and $A(\pi(x))$, where $\pi(x) = (\pi(t_1), \dots, \pi(t_L))$ permutes token identities while preserving absolute positions. The permutation sensitivity is:

$$PS_h = \mathbb{E}_{x, \pi} [\|A_h(x) - A_h(\pi(x))\|_F]$$

Validation. On the entropy=0 Markov model: `pos_dep` classifies 16/16 heads as positional; PS classifies 16/16 heads as positional (PS < 0.01 for all heads). The two metrics agree on ground truth.

On Shakespeare (depth 6): `pos_dep` classifies 47/48 heads as positional. PS classifies 9/48 as positional, resolving 39 false positives. All 39 false-positive heads are in layers 1–5 (not layer 0). The failure mode: by layer 1, the residual stream already encodes rich content information, and content-conditional attention produces patterns with low cross-input variance.

Figure 6: Permutation Sensitivity Metric Validation

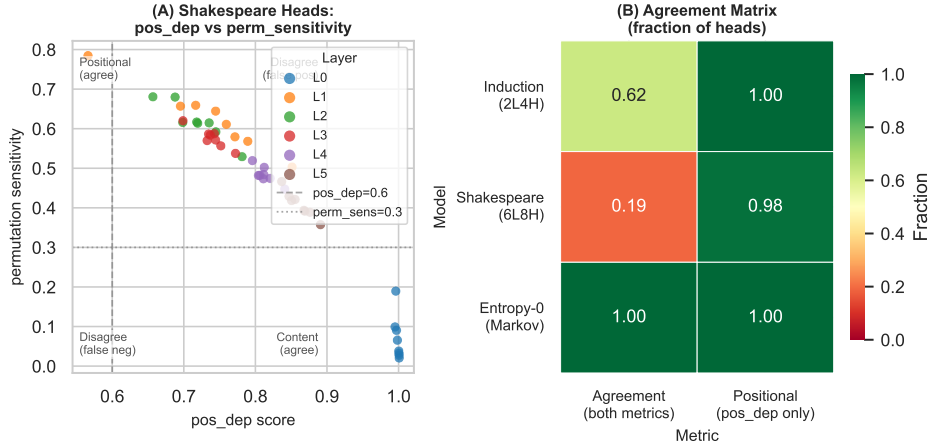


Figure 7. Permutation-sensitivity metric validation on Shakespeare. Scatter of `pos_dep` vs. permutation sensitivity PS for all 48 attention heads in the depth-6 Shakespeare model. Only 9 heads are classified as positional by both metrics (bottom-left quadrant); 39 heads have high `pos_dep` but high PS (false positives under the old metric: content-based but pattern-stable). These 39 are concentrated in layers 1–5; Layer 0 (blue) heads largely agree.

Table 14. Critical-head ablation across 4 seeds of the content-matching induction model. All critical heads are in Layer 0; specific head identity varies.

Seed	Baseline accuracy	Critical head	Accuracy after ablation	Drop
0	0.854	L0H3	0.041	0.813
1	0.964	L0H1	0.118	0.846
2	0.965	L0H1	0.028	0.937
3	0.968	L0H3	0.463	0.505
Mean drop				0.775 ± 0.162

Table 15. Statistical summary of key claims.

Claim	Point estimate	95% CI	Null	<i>p</i> -value
Layer-level agreement (20 seeds)	100%	[100%, 100%]	50%	< 0.001
Head-level agreement (20 seeds)	23.2%	[17.4%, 29.5%]	12.5%	0.002
Head-level similarity (20 seeds)	0.329	[0.28, 0.38]	0	< 0.001
Ablation drop (4 seeds)	0.775	[0.59, 0.91]	0	0.004
Suppression slope (n=4 depths)	0.059/layer	[0.023, 0.109]	0	0.052

C Full Results Tables

C.1 Ablation robustness: four seeds

C.2 Bootstrap confidence intervals

D Ablation Protocols

Zero-ablation. We zero-out a component’s output by setting its contribution to the residual stream to the zero vector. This is the least assumption-dependent ablation: it tests whether the component’s *contribution* is necessary, without assuming a particular “baseline” activation.

Critical-head identification. For each model, we identify the critical head as

$$h^* = \arg \max_h (\text{accuracy}_{\text{baseline}} - \text{accuracy}_{h\text{-ablated}}).$$

We verify that this single head accounts for the majority of the model’s induction ability by checking that the remaining accuracy after ablation is near the ablated-model floor (≈ 0.09 – 0.12 for 2L/4H content-matching models), far below the functional baseline of ≈ 0.96 .

Transplantation ablation. In the clean-slate transplant condition, we zero the recipient’s critical head before inserting the donor head. This tests whether the donor head can function even without the recipient’s critical-head initialization (ruling out residual activation from the original weights).